

TOWER OP RE-BENCHMARK #2 - post-#2713 / #2805

Heege Tower OP print set - 12 sha256-verified photos - REAL Telegram print-translator path - 2026-07-19

Run: CHARLIE worktree @ main 2ff4e78e1 (v3.174.0 - full overnight train #2800/#2801/#2802/#2804/#2805/#2713/#2806 + #2803), this PR's parents[4] driver, Doppler stg scoped 5-var process-env only, \$0.00 total inference (every LLM_CALL = together/gemma-3n-E4B-it; OpenAI interpreter 429 quota-dead, identical to both priors). Judged per pack protocol: 12 independent adversarial vision judges, blind to prior verdicts, + 2 full-reply corrections.

VERDICT

- 1. **#2713 DELIVERED - THE ROUTING WALL IS BROKEN.** 4 of the 5 cases that died before vision in both prior runs (c06, c08 caption-gate; c11, c12 table misclass) now enter the pipeline and produce real answers. 11/12 reach vision classification (was 7/12).
- 2. **#2805 DELIVERED DETECTION, NOT PREVENTION.** The new false_absence_claim autoeval rule fired live 4x in-run (P1 on c01/c07; inside P0s on c06/c09) - the exact class that had no rule before. The prompt-side contract did not change gemma's behavior.
- 3. **SYNTHESIS IS NOW THE ISOLATED BOTTLENECK.** Corrected mean 1.0/10 - exactly flat vs rerun1 - with 10/12 cases graded on real answers instead of 7/12. Every remaining point is owned by the free vision model's synthesis: fabricated tags, wrong-row lookups, laundered OCR. This is the strongest empirical case for PR-F yet.

Scorecard (three runs, same photos, same protocol)

Run	Mean /10	Corrected	Clean	Reach vision	Cost	Notes
Baseline 07-18	1.7	-	0/12	7/12	\$0	evidence spine inert (0 OCR items everywhere)
Rerun1 (post-#2800)	1.0	-	0/12	7/12	\$0	evidence flows (18-75 items) but answers flat routing fixed; harness split-reply artifact
RERUN2 (this run)	0.5	1.0	0/12	11/12	\$0	corrected; judges punished wrong-row lookups hard

The corrected mean applies the two full-reply re-grades (c06, c09) after discovering final_text captured only the last Telegram chunk of split replies (see H-fixes). Protocol-lane numbers kept for comparability.

PER-CASE (baseline -> rerun1 -> RERUN2)

Case	Base	R1	R2	What happened this run
c01 K1 coords	F 1	F 1	F 0	Never names K1; generic theory template; invents F3, S1-S5 as 'Evidence'. Autoeval P1 fired.
c02 motor ratings	F 1	F 0	F 0 honest	Theory call lost at Together 30s timeout -> honest fallback. ZERO fabrication (rerun1: ~236 invented tags). 2.2kW/4.95A absent from OCR (small font).
c03 K4.1-4.4 role	P 5.5	P 4	F 2	Misses printed 'torque limitation (R1)/braking relay' headers; launders '12-G122' (real: Q1.2-Q12.2); invents F4.
c04 TDC switch	F 2	F 1	F 0 NEW DROP	handled=false, no classify: wiring carve-out claims the caption pre-vision (L1). Judge's correct answer: S7.1 -> I6.1 @ X5.2.3.
c05 sensor P/Ns	F 2	F 0	F 0 honest	Theory timeout -> honest fallback. Garbled fragments of BOTH correct part numbers sit in its own ocr_items (xSi-MERHO <- XS1-N18PC410).
c06 pawl switches	0 gate	0 gate	F 0/1 ROUTES NOW (86 items). Reply fabricates pawl attribution + K1-K96 ANSWERS enumeration; S21/S22 genuinely NOT on this sheet -> honest absence was the right answer (L7). P0 fired.	
c07 S19 meaning	F 1	F 1	F 1	False-absence persists ('not explicitly labeled' under the printed 'rope control' header); invents K1/F1/Q1. P1 caught it in-run.
c08 pretension relays	0 gate	0 gate	F 2 ROUTES NOW (53 items). Reads header + car columns correctly, then fabricates K1-K4 for the printed K5.1-K5.4; self-contradicts.	
c09 supply feeds	P 6	P 5	F 0 / P 5 full	Main chunk had the CORRECT 480V+240V answer verbatim - buried in ~80 fabricated tags across 4 degenerate enumerations; tail chunk (graded lane) = laundered glare-OCR. P0 fired.
c10 FF LED	0 gate	0 gate	F 0 mutated	Caption gate now passes; vision says NAMEPLATE 0.67 (unit-vocab V/Hz native to LED tables) -> declined. 170 OCR items incl. answer fragments (L2).
c11 IG 1Hz flash	0 misclass	0 misclass	F 1 ROUTES NOW (184 items). Wrong-row lookup: quotes X4.3 LED1's row for ANSWERS X4.4 LED5; the correct row IS in its own OCR.	
c12 X6.3 elem 5-8	F 0	0 misclass	F 0 ROUTES NOW (156 items). Cross-module substitution: real X6.1/X6.2 text ANSWERS presented as X6.3; 0/4 on a safety-interlock table.	

PER-CLASS (the comparison that matters)

Failure class	Base	R1	R2	Call
Caption-gate drops (R1)	c06 c08 c10	same 3	0 of class	FIXED as a class; c10 re-drops on NAMEPLATE (mutated)
Table-page misclass (R2)	c11 c12	same 2	0	FIXED - both answer at 0.80/0.76 conf
Residual/new routing	-	-	c10, c04	NEW: NAMEPLATE escape (L2) + wiring carve-out (L1)
Fabricated device tags	c02 c09	c01 c02 c07 c09	c01 c03 c06 c07 c08 c11	PERSISTS - but the 2 worst floods now die honestly at timeout
'Not labeled' false absence	c05 c07	c05 c07 c09	c07 (+c06 full)	REDUCED surface; now DETECTED live by the #2805 rule
Garbled-OCR laundering	-	c03 c05 c09	c03 c09	PERSISTS (prompt contract doesn't stop it)
Degenerate enumeration	c02	c02 c09	c06 c09 (full)	PERSISTS, moved cases; both P0-flagged live
Wrong coordinate convention	c01	c01	c01	PERSISTS/WORSE (question ignored entirely)
Wrong-row / cross-module lookup	-	-	c11 c12	NEW CLASS unmasked by routing fix - confident wrong safety facts; invisible to false_absence_claim (L5)
Theory-timeout honest fallback	-	-	c02 c05	NEW CLASS - 30s Together ceiling; honest > fabrication but 2/10 answers lost (L3)

NEXT ROUND - CONFIRMED LEVERS (code-verified, not hypotheses)

Every mechanism below was verified against main 2ff4e78e1 on 2026-07-19: the run's evidence names the failure, the code names the cause. Ordered by effect per unit of work. L1-L4 alone make round 3 a 12/12, honestly-measured, pure synthesis benchmark.

L1 TRIVIAL FIXED Narrow the wiring carve-out (restores c04)

Mechanism (verified): bot.py:1140 `_try_print_translator_reply` declines when `wiring_intake.parse_wiring_intent(caption).kind != 'none'` (line 1177). The parser returns THREE kinds (`intake | question | none`, `wiring_intake.py:246-258`); the carve-out was written for intake ownership (its own comment says 'an explicit wiring-intake command') but also fires on 'question'. c04's caption ('...which PLC input is the switch WIRED TO...') parses as a wiring question -> print rung declines pre-vision (capture: `handled=false`, no classification - first time ever; both priors answered c04).

Fix: Decline only on `kind == 'intake'`. Regression test: c04's caption verbatim + print photo must classify.

Effect: Wiring-phrased print questions - a large share of real technician phrasing - stop dying silently. +1 case enters.

L2 SMALL FIXED Dense-table override must outrank NAMEPLATE unit-vocabulary (restores c10)

Mechanism (verified): `vision_worker.py: NAMEPLATE_OCR_FIELDS` (≥ 3 unit fields - `hp/kW/Hz/volts` - in `OCR => NAMEPLATE` 'regardless of vision description', ~line 103) is checked BEFORE `DENSE_TABLE_OCR_THRESHOLD=50` (line 685). c10's LED-reference page carries '24 V', '10 Hz', '(25 Hz)', 'voltage' as `NATIVE` table content -> `NAMEPLATE` 0.67 with ~170 OCR items; the dense override never ran. #2713's adversarial probes protect real dense nameplates via exactly this precedence.

Fix: Ratio-aware, not a blind reorder: at ≥ 50 items, holding `NAMEPLATE` requires plate vocabulary proper (`NAMEPLATE_KEYWORDS`) or a unit-field `RATIO` (3 hits among 12 items = plate; 4 among 170 = table noise). Fixture: c10's capture from this run.

Effect: With L1+L2: 12/12 pipeline entry. The last original routing survivor closes.

L3 KNOB TRIVIAL Lift the Together vision timeout (stops losing computed answers)

Mechanism (verified): `router.py:209` - the together provider hardcodes `timeout=30.0` (`dataclass` default is 60.0 at line 152; no env knob). Vision/theory calls are Together-only - NO cascade fallback exists for vision. This run's successful theory calls measured 13.9-28.6s; 2 of 10 (c02/c05) crossed 30s -> 'together timeout after 30s' -> 'All providers exhausted' -> honest fallback. #2804's 2000-token budget + #2805's larger prompt consumed the old margin.

Fix: `TOGETHERAI_TIMEOUT` env knob (default ~90s for the vision path) + compose mapping + docs/env-vars.md row (the enumerated compose env-block trap has bitten twice before).

Effect: Recovers the 2-in-10 lost-answer class. `SLOW_LLM_CALL` warnings become the early ceiling indicator.

L4 PACK SMALL Bench capture fixes (measurement honesty)

Mechanism (verified): (a) #2804-length replies now split at Telegram's limit; `final_text` captured only the TAIL chunk - c09's correct 480V/240V answer sat ungraded in the main chunk (laneA captures verified single-chunk: this is new drift). (b) c04 and c10 were indistinguishable in-capture (both `handled=false`, no reason). (c) provider/model fields record interpreter config, not the answering provider - misled judges in two consecutive runs.

Fix: `final_text` := ordered concatenation of reply chunks (minus ack); add a `decline_reason` field; record the actual answering provider from the run's `LLM_CALL` telemetry.

Effect: The bench grades what the technician actually saw. No more silent under-measurement of split replies.

L5 RULE NEW New autoeval rule: wrong-row / cross-module table lookup

Mechanism (verified): c11 answered X4.3 LED1's row for X4.4 LED5; c12 presented X6.1/X6.2 row text as X6.3 elements 5-8 (0/4 on a safety-interlock table). Both quote `VERBATIM-REAL` print text from the `WRONG` cell - `false_absence_claim` structurally cannot see it (nothing is claimed absent), and it is worse than absence: confident wrong safety facts.

Fix: Deterministic detection is feasible: the question names module+element; `ocr_tokens` carries line-grouped word boxes (`{text, bbox, line}`) - when the reply's quoted row text matches a different module's line than asked, flag P1. Fixtures: c11/c12 captures from this run.

Effect: The scariest new class gets a tripwire before any customer sees it.

L6 MIKE PR-F paid interpreter - the empirical case is now clean

Mechanism (verified): With routing repaired and evidence attached, the free VL floor (gemma-3n-E4B - the ONLY serverless vision model this Together account can reach) scores 0.5-1.0/10 on synthesis alone: reads structure (c08's header+columns) then fabricates the tags beneath; drops its own correct detections; cannot do row-accurate lookups. c09-fullreply proves correct extraction happens and then drowns in degenerate enumeration.

Fix: Blocked on: credits + printsense/interpret.py sign-off (never-calibrate guarded file). Run round 3 free-lane vs paid-lane side by side under the existing \$-bounded bench guard.

Effect: Directly attacks the fabrication, wrong-row and laundering classes that own every remaining point.

L7 TRUTH PACK Truth-set maintenance: c06's pass condition = honest absence + redirect

Mechanism (verified): Judge verified against the photo: S21.x/S22.x pawl switches are NOT on c06's sheet (that page groups car No.3/No.4 inputs; the pawl tags live on the X6.3 table - c12's page). cases.json's expected implies findability; the only correct c06 answer is honest absence plus a redirect to the right sheet.

Fix: Correct the case's expected per the never-weaken-truth discipline (primary-source verification is recorded in the verdict).

Effect: Round 3's c06 rewards honesty instead of demanding an impossible lookup.

WHAT ROUND 3 LOOKS LIKE IF L1-L4 LAND (zero model change)

12/12 pipeline entry - ~12 graded real answers - honest full-reply measurement. The bench becomes a pure synthesis benchmark: every remaining point is the model, nothing else. L5 guards the scariest class; L6 (PR-F) is what moves the score. Structural wins already banked this round: OCR evidence 18-184 items on every classified case with tesseract provenance; classification confidence 0.76-0.95; the observability regime caught real garbage live (2x P0 + 2x P1, including the brand-new false_absence_claim rule on exactly its target class); honest failure now exists as a floor where fabrication floods used to be; \$0.00 end-to-end, three runs in a row.

Evidence & artifacts

Committed: tools/internet_print_test/benchmarks/2026-07-18-towerop/RERUN2-2026-07-19-post-2713-2805.md (PR #2808, b30b7b55d)

Priors: REPORT.md (baseline 1.7) + RERUN-2026-07-19-post-2800.md (rerun1 1.0), same pack

Run truth: CHARLIE ~/towerop/out-0719/*.json + ~/towerop/run2.log (LLM_CALL / PRINT_AUTOEVAL lines)

Judging: 12 blind adversarial sonnet verdicts + 2 full-reply re-grades (session archive)

Deploys: prod green through 2ff4e78e1 (six deploy-vps successes overnight); staging 3.173.0 verified